

Yash Jadhav

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EDUCATION

Syracuse University, College of Eng & Computer Science

Syracuse, NY

Master of Science in Computer Science

August 2024 – May 2026

Relevant Coursework: Data Structures, Computer Architecture, Design and Analysis of Algorithms, Assured Programming with Formal Methods, Principles of Operating Systems, Social Media & Data Mining.

University of Mumbai, Vidyalkar Institute of Technology

Mumbai, Maharashtra

Bachelor of Science in Electronics & Telecommunication

June 2020 – May 2024

Minor: AI & ML.

Relevant Coursework: Cloud Computing & Security, Web Design, Control Systems, Database Management Systems, Natural Language Processing, Computer Vision, Probability & Statistics, Deep Learning, Data Science & Analytics.

WORK EXPERIENCE

Syracuse University

Syracuse, NY

Graduate Research Assistant

May 2025 – August 2025

- Enhanced ARIES compiler for PIM architectures on Xilinx VCK190 FPGAs, reducing ML workload latency by 40%.
- Engineered custom MLIR passes to optimize GEMM and MTTKRP kernels, boosting throughput by 2x.
- Devised automated testing pipelines with Python and Bash, cutting debugging cycles by 25%.
- Authored to research deliverables and open-source GitHub repos leveraged by 30+ collaborators.

TIH-IoT - IIT Bombay

Mumbai, India

Project Fellow

April 2023 – January 2024

- Devised and deployed IoT data pipelines using Python and SQL to collect and process sensor data, enabling real-time monitoring of 1,000+ records.
- Developed anomaly detection workflows and backend scripts that improved data processing accuracy by 22%.
- Automated data ingestion, transformation, and reporting workflows with MATLAB + REST APIs, diminishing manual effort by 40%.
- Integrated sensor outputs into centralized databases, supporting data-driven decision-making for research teams.
- Collaborated in a cross-functional team of 5, delivering a scalable prototype adopted as a reference project for TIH-IoT.

TECHNICAL SKILLS & CERTIFICATIONS

- Technical Skills: Python, Java, SQL (PostgreSQL, MySQL, MongoDB); Node.js, Express.js, Flask, REST APIs, JWT, OAuth2; AWS (EC2, S3), Docker, GitHub Actions, CI/CD, Postman; Pandas, NumPy, ETL/ELT, Apache Airflow; TensorFlow, PyTorch, Hugging Face, LangChain, BERT; Git, Jupyter, Jira, Tableau, Power BI.
- Cloud Computing – by University of Illinois
- IBM Back-End Developer Professional Certificate

PROJECTS

Banking System

March 2025 – June 2025

- Built a secure and scalable backend system simulating real-time transactions and automated receipt generation using Stripe API.
- Designed and deployed 10+ RESTful API endpoints with JWT-based authentication and role-based access control to ensure secure login and transaction validation.
- Architected a normalized MySQL database with 5 relational tables for users, accounts, balances, and transactions, improving data integrity and query efficiency by 30%.

E-Commerce Backend Platform

October 2024 – January 2025

- Developed a modular backend handling user accounts, product listings, carts, orders, and checkout workflows.
- Implemented and documented 15+ RESTful API endpoints for authentication, product CRUD, order tracking, and admin role management.
- Integrated Stripe API for secure payment workflows with automated receipt generation, enhancing transaction handling and customer trust.
- Applied JWT authentication and session management, reducing unauthorized access risks and strengthening platform security.